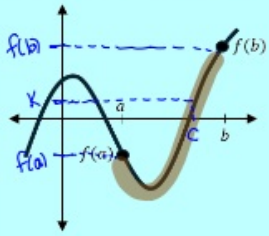
CHA-4.B.1 The average value of a continuous function f over an interval $[a, b]$ is $\frac{1}{b-a}\int_a^b f(x)dx$.

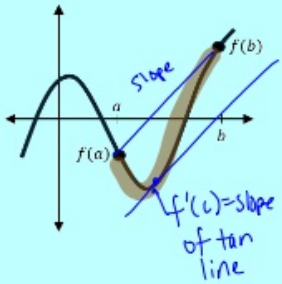
8-1 Average Value of a Function

Get Ready...

Refresh your memory about the following theorems/topics:

Intermediate Value Theorem (IVT)	**If** f is continuous on the closed interval $[a, b]$, where $f(a) \neq f(b)$ and $f(a) < k < f(b)$, **then** there is at least one number c in $[a, b]$ such that $f(c) = k$
Average Rate of Change	Given a function $f(x)$ and interval $[a, b]$: $AROC = \dfrac{f(b) - f(a)}{b - a}$
Mean Value Theorem (MVT)	**If** f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b), **then** there exists a number c in (a, b) such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ (or IROC = AROC somewhere in the interval)





Average Value of a Function	If $f(x)$ is continuous on the closed interval $[a, b]$, then the average value of the function over the interval is: $$\text{Ave Value} = \frac{1}{b-a}\int_a^b f(x)\,dx$$

The Mean Value Theorem for Integrals	If $f(x)$ is continuous on the closed interval $[a, b]$, then there exists a number c in $[a, b]$ such that: $$f(c) = \underbrace{\frac{1}{b-a}\int_a^b f(x)\,dx}_{\text{Ave value}}$$ or equivalently: $\int_a^b f(x)\,dx = f(c)(b-a)$

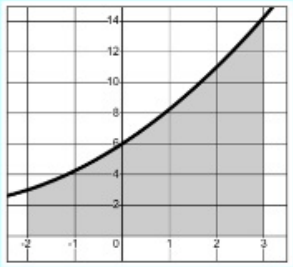
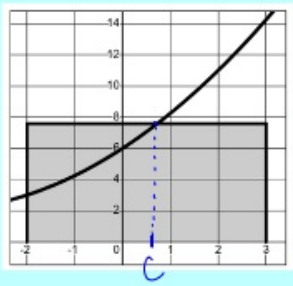
Ave Value of $f(x)$ over $[a,b]$ $= \frac{1}{b-a}\int_a^b f(x)\,dx$

$$= \frac{\int_a^b f(x)\,dx}{b-a}$$

What does this: $\int_a^b f(x)\,dx = f(c)(b-a)$ mean geometrically?

height base

It means there is a rectangle with width of $(b-a)$ with the same area as the area under the curve:

The graph of $f(x) = 0.25x^2 + 2x + 6$ is shown below. The area under $f(x)$, above the x-axis, and between $x = -2$ and $x = 3$ is calculated to be approximately 37.917. 	A value c exists between $x = -2$ and $x = 3$ such that $f(c)(b-a)$ equals the area under the curve $f(x)$. Note: The value of $c \approx 0.73$ and $f(0.73) \approx 7.5833$ Area $\approx 7.5833(3-(-2)) = 37.917$ 

Example 1: Given $f(x) = 4 - x^2$ on the interval $[0, 3]$.
$a \quad b$

(a) Find the average value of $f(x)$ on the given interval.

$$\text{Ave value of } f = \frac{1}{b-a}\int_a^b f(x)\,dx$$

$$= \frac{1}{3-0}\int_0^3 [4-x^2]\,dx = \frac{1}{3}\left[4x - \frac{1}{3}x^3\right]\Big|_0^3$$

$$= \frac{1}{3}\left[12 - 9 - (0-0)\right] = \frac{1}{3}(3) = \boxed{1}$$

(b) Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x)$ on the given interval.

$$a \le c \le b \qquad f(c) = \underbrace{\frac{1}{b-a}\int_a^b f(x)\,dx}$$

$$\left\{\begin{array}{l} f(x) = 4 - x^2 \\ f(c) = 4 - c^2 \end{array}\right.$$

$$4 - c^2 = 1$$
$$-4 \qquad -4$$
$$-c^2 = -3$$
$$c^2 = 3$$
$$c = \pm\sqrt{3}$$

$-\sqrt{3}$ is not in $[0,3]$

$$0 \le c \le 3$$

$$\boxed{c = \sqrt{3}}$$

Example 2: A research team is studying the concentration of a medication in a patient's bloodstream over time. The concentration (in mg/L) is modeled by the function:

$$C(t) = 5e^{-0.4t} + 0.8\ln(t+1)$$

where t is measured in hours, and $0 \le t \le 6$.

(a) What is the average concentration of the medication in the bloodstream during the first 6 hours? Show the setup that leads to your answer. Ave value of $C(t)$ over $[0,6]$

Ave concentration of medication $= \frac{1}{6-0}\int_0^6 C(t)\,dt$

$= 2.910520403\ldots$

≈ 2.911 mg/L

(b) When will the concentration of the medication in the bloodstream equal the average concentration of the medication in the bloodstream? Show the setup that leads to your answer.

$\underbrace{C(t)}_{\text{in } Y_1} = \underbrace{2.910520403\ldots}_{\text{in } Y_2}$ on calc.

$t = 2.3681151\ldots$

$t = 2.368$ hours.